

# Pragnosia: Making a Diagonal-Complex Spin Recurrence the Load-Bearing Core of a Language Model

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**Abstract**—We study a transformer/state-space hybrid in which a rotational “spin” recurrence, a diagonal-complex linear-recurrent unit run as a parallel associative scan, is the *core token-mixer*, with attention only a periodic helper. The work has a sharp, honest arc. We first scaled the hybrid the wrong way: a single spin carrier placed after a transformer stack. Its parameter share collapses  $5.76\% \rightarrow 1.90\% \rightarrow 0.37\%$  from 21M to 1.4B, and at 1.4B a causal probe shows the carrier contributes only 0.028% of the loss, vestigial as built. A parameter-matched ablation then shows the intended design. With spin as the mixer, validation perplexity is 219 vs 279 for an attention-only baseline at matched parameters, a quality win when spin is the core. A control that swaps the carrier for a real, no-phase recurrence in the same placement also beats attention ( $3.5\times$ ), so at the tested budget the effect is about placement, not phase. Separately, we establish the design’s core property at scale as causal attribution. In a 284M spin-dominant model (validation perplexity 24.54 on a 176.6B-token corpus), ablating the spin carrier sends perplexity  $25 \rightarrow 7716$  ( $306\times$ ), while ablating attention costs only  $3.4\times$ : the carrier is the load-bearing core, the exact inverse of the 1.4B drift. We keep the two claims distinct: the 284M result shows the carrier is used, not that the design wins; a matched attention-only transformer at 284M is the key experiment still to run. On this substrate runs a proposed self-calibrating controller (mechanisms described; validation preliminary): surprise-modulated continual learning with generative replay, a persistent in-weights episodic memory, an introspection layer, and function-preserving self-growth on probe-confirmed saturation, with no hand-set calibration thresholds. We report the limits plainly: the spin-dominant win is small-scale and single-seed, the grown models are undertrained, and the honesty signal does not yet separate knowledge from confabulation at 284M.

**Index Terms**—State-space models, linear recurrence, sequence modeling, language models, causal ablation, continual learning.

## I. INTRODUCTION

THE dominant paradigm in AI is the large language model: a transformer [1] trained to predict the next token. Language models are capable but, relative to the goal of an adaptive, persistent learner, three properties are missing by construction. They are *frozen*: once pretraining ends they cannot acquire a fact without fine-tuning or prompt-stuffing, and naive updates cause catastrophic forgetting [2]. They are *over-confident*: lacking a model of their own uncertainty, they fabricate rather than abstain. And their computation is opaque: there is no standard intervention separating in-state computation from memorized retrieval.

Manuscript submitted for double-blind peer review. Author names, affiliations, and contact information are withheld.

TABLE I

SUMMARY OF THE CENTRAL RESULTS. THESE ARE TWO CLAIMS, KEPT DISTINCT: A SMALL-SCALE *quality* WIN (SPIN-DOMINANT BEATS A MATCHED TRANSFORMER) AND A LARGE-SCALE *causal-dominance* PROPERTY (THE CARRIER IS LOAD-BEARING). THE 284M ROWS SHOW THE CARRIER IS USED, NOT THAT THE DESIGN WINS AT SCALE.

| Result                      | Value                       | Shows                              |
|-----------------------------|-----------------------------|------------------------------------|
| Carrier share, 21M–1.4B     | $5.76\% \rightarrow 0.37\%$ | a fixed carrier dilutes with scale |
| Carrier causal signal, 1.4B | 0.028% of loss              | vestigial as built                 |
| Param-matched abl. (37M)    | 219 vs 279                  | spin-dominant by 21%               |
| 284M spin-dominant          | val ppl 24.54               | intended build, real scale         |
| 284M carrier ablation       | $306\times$ vs $3.4\times$  | carrier is load-bearing            |
| 1.4B “wrong build”          | val ppl 17.35               | bigger, yet vestigial              |

Pragnosia is an attempt to build the missing properties into the substrate itself, even at small scale. Its contributions are: (i) a diagonal-complex rotational recurrence used as the *core token-mixer* via a parallel scan (Section III); (ii) a measured analysis of *where that carrier earns its parameters*: our first scaled model drifted to attention-dominant with a causally-negligible carrier, while a parameter-matched ablation shows the intended *spin-dominant* design wins by about 21%, is confirmed load-bearing at 284M by ablation, and generalizes to a different (real, no-phase) recurrence in the same placement (Section IV); (iii) a self-calibrating controller with surprise-modulated continual learning, a persistent in-weights episodic memory, an introspection layer, and function-preserving self-growth, with no hand-set constants (Section V); and (iv) an honest account of what works and what does not (Sections VI–VII). We release the system and every measured number.

## II. RELATED WORK

**Recurrent and state-space sequence models.** The carrier is a diagonal linear recurrence in the lineage of structured state-space models: S4 [3] and its diagonal variants, the Linear Recurrent Unit (LRU) [4], and S5; and of recent recurrence-as-mixer models: Mamba [5], RWKV [6], RetNet [7], and the linear-attention and xLSTM families. Our carrier itself is not novel; it is a standard diagonal-complex LRU run as a parallel associative scan.

**Where we differ: placement and verification.** These models are typically the sole mixer, or interleaved uniformly

with attention. Our question is narrower: in a hybrid, is the recurrence carrying the computation, or is it a vestigial side-channel? We contribute (i) a measured account that a single carrier’s contribution collapses with scale when placed after a transformer stack, and (ii) a causal protocol (carried-state intervention on single-carrier models, direct ablation on the spin-dominant design) to test whether the mixer is load-bearing. Our headline ( $306\times$  ablation cost vs  $3.4\times$  for attention at 284M) is a property of placement, not of the recurrence.

**Continual learning and memory.** The controller’s no-forgetting mechanism is generative replay / pseudo-rehearsal [8]; its surprise-gated learning rate is in the spirit of neuromodulation [9]; its in-weights episodic store is a fast-weight associative memory (related to modern Hopfield networks [10]), not retrieval-augmented generation. The contribution is their integration into one self-calibrating model with no hand-set constants.

### III. THE SPIN CARRIER

#### A. A diagonal-complex recurrence, run as a parallel scan

The carrier is a linear-recurrent unit [3] over a complex hidden state. For input  $x$  (layer-normed to  $y$ ), with an input-dependent gate  $g = \sigma(Gy)$ , the per-channel recurrence and read-out are

$$b_t = g \odot (U_{\text{re}} y_t + i \cdot U_{\text{im}} y_t), \quad h_t = \lambda \odot h_{t-1} + b_t, \quad (1)$$

$$\lambda_j = e^{-\exp(\nu_j)} \cdot e^{i\theta_j}, \quad \text{out}_t = C[\text{Re } h_t; \text{Im } h_t], \quad x_t += \sigma(\gamma) \text{out}_t$$

Each channel  $j$  is a *damped complex oscillator* at its own frequency  $\theta_j$  and decay  $|\lambda_j| \in (0, 1)$ : information is carried in the evolving phase of the state, read back to the real stream through  $C$  and added by a learned gate  $\sigma(\gamma)$ . Because  $\lambda$  is diagonal the recurrence is matmul-free,  $O(Td)$  rather than  $O(Td^2)$ , and associative, so it is computed not by a  $T$ -step loop but by a log-depth parallel associative scan (a Hillis–Steele scan). An in-place variant is exact and about 20% faster on the forward pass but breaks autograd; we keep the materializing scan.

#### B. Why phase

Encoding state in the phase of a damped oscillator has three motivations. (i) *Capacity without interference*: a bank of oscillators at distinct frequencies represents many independent slowly-varying quantities side by side, much as Fourier components are orthogonal. (ii) *Stable, multi-timescale memory*: with  $|\lambda| < 1$  the state decays gracefully, and the per-channel decay  $\nu$  is learned, so channels hold information over different horizons. (iii) *Parallelizable dynamics*: a diagonal-complex recurrence is associative, which admits the log-depth parallel scan that makes training practical, a property a nonlinear recurrence lacks. We do not claim phase is necessary; the empirical question is whether such a mixer, placed as the core, carries the computation.

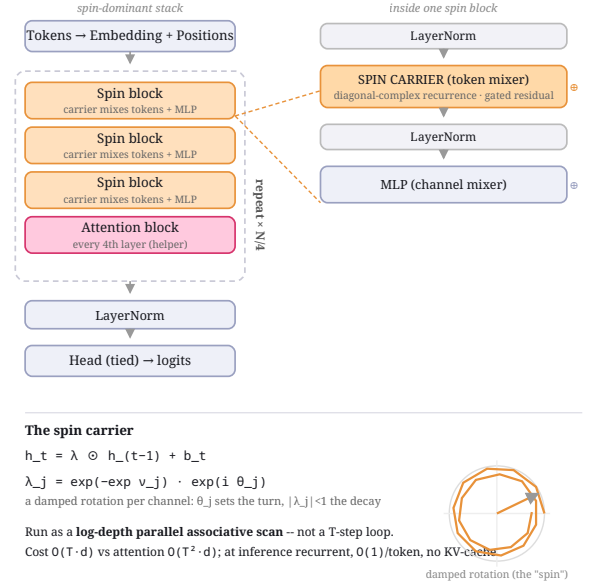


Fig. 1. The spin-dominant architecture. **Left**: the macro stack; the spin carrier is the token-mixer in three of every four blocks, with an attention block every fourth. **Right**: one spin block. **Bottom**: the carrier is a per-channel damped complex recurrence  $\lambda_j = e^{-\exp \nu_j} e^{i\theta_j}$  evaluated by a log-depth parallel associative scan,  $O(Td)$  and recurrent  $O(1)/\text{token}$  at inference.

#### C. Architecture and modes

A SpinBlock uses the carrier as its token mixer (a strong gate,  $\sigma(\gamma) \approx 0.88$ ) followed by an MLP; a standard attention Block is the alternative. The model exposes one config flag,  $\text{carrier} \in \{\text{none}, \text{single}, \text{per\_block}, \text{spin\_dominant}\}$ . *none* is a transformer; *single* places one carrier after the stack; *spin\_dominant*, the intended design, makes the carrier the mixer in 3 of every 4 layers, with attention every 4th as a helper. At long context the carrier is  $O(Td)$  versus attention’s  $O(T^2d)$ , and at inference it is recurrent:  $O(1)$  per token, constant memory, no KV-cache growth. Fusing the scan gives about  $2\times$  training throughput ( $49\text{K} \rightarrow 100\text{K}$  tok/s, batch 24) on a consumer GPU. It is not faster than attention on short (256-token) training sequences; its intended wins are long context and recurrent inference.

### IV. WHERE THE SPIN CARRIER EARNS ITS PARAMETERS

#### A. The scaled model drifted to attention-dominant

Our first scalable design placed a single gated spin carrier after a transformer stack. A single carrier is one fixed  $d$ -shaped layer (about 5.25M params), so as the model grows by depth and width its parameter share collapses:  $5.76\%$  (21M)  $\rightarrow 1.90\%$  (276M)  $\rightarrow 0.37\%$  (1.4B). We measured its causal contribution by a carried-state intervention on the single-carrier model (transplant the carrier state across a segment boundary): at 1.4B the ordering own  $<$  swapped  $<$  random still holds directionally, but the contribution is only 0.00086 NLL, 0.028% of the loss, down about  $20\times$  from 276M. As built, the carrier is causally negligible at scale: the model is a transformer with a vestigial side-channel. The same 1.4B

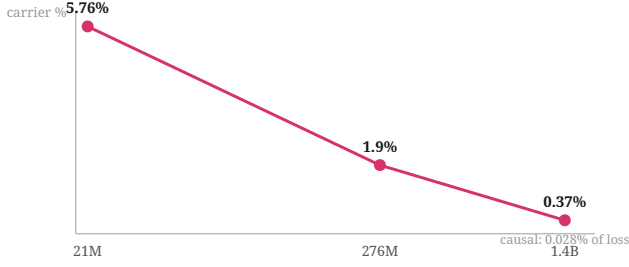


Fig. 2. A fixed single spin carrier is one  $d$ -shaped layer, so as depth and width grow its parameter share collapses (5.76%  $\rightarrow$  1.90%  $\rightarrow$  0.37%) and its causal contribution falls to 0.028% of the loss at 1.4B, the motivation for making the carrier the core.

TABLE II  
PARAMETER-MATCHED ABLATION. SPIN AS THE TOKEN MIXER IS THE MOST PARAMETER-EFFICIENT.

| Design                                    | Params | Val ppl      |
|-------------------------------------------|--------|--------------|
| Attention-only (transformer)              | 35.7M  | 279.0        |
| <b>Spin-dominant (intended)</b>           | 37.3M  | <b>219.1</b> |
| Per-block hybrid (attn + spin each block) | 46.2M  | 228.0        |
| Param-matched transformer-only (11L)      | 45M    | 270          |

run also over-grew: an earlier trigger grew on any validation plateau rather than true saturation, so an undertrained model ballooned to 1.4B long before the tokens needed to fill it. Section V gives the corrected growth rule.

#### B. The intended design is spin-dominant, and wins at matched parameters

The drift inverts the intended architecture, where spin is the core mixer. We test the intended design with a clean parameter-matched ablation (same corpus, same budget,  $d = 512$ , single seed): Table II and Fig. 3.

Spin-dominant wins by about 21% over the attention-only baseline at matched parameters and beats the larger per-block hybrid and matched transformer with fewer weights. The carrier earns its parameters when it is the core mixer, not a side-channel. This ablation is small-scale and short-budget (though the core comparison is now confirmed seed-robust, next) and must be replicated at 200M–1B before it is proven.

#### C. Seed robustness

To check the result is not an artifact of one initialization or of parameter count, we re-ran the comparison across three seeds for four architectures, each at a matched short budget (about 30M tokens) with a per-architecture learning rate set by a Smith range-test [11]. The ordering is highly reproducible (Table III).

Two readings. (i) The gap is not seed noise, nor capacity: the standard deviation is 1–2% of the mean, and neither a larger transformer (11L, 45M) nor a side-channel carrier (46M) escapes the attention-family band (329–361) despite more parameters than spin-dominant (37M). The win is placement, not parameter count. (ii) The magnitude is budget- and

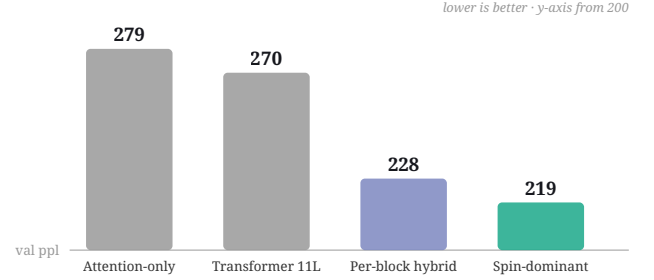


Fig. 3. Parameter-matched ablation (37–46M, same corpus and budget, single seed). Spin as the token-mixer (219) beats the attention-only baseline (279) by about 21%, and the larger per-block hybrid (228) and matched transformer (270), with fewer or equal weights.

TABLE III  
MULTI-SEED ROBUSTNESS ACROSS THE ATTENTION FAMILY VS SPIN-AS-CORE (3 SEEDS EACH; PER-ARCHITECTURE RANGE-TEST LR; MATCHED 30M-TOKEN BUDGET).

| Architecture                        | Params | lr     | Val ppl (mean $\pm$ std)          |
|-------------------------------------|--------|--------|-----------------------------------|
| Attention-only (8L)                 | 35.7M  | 1.0e−4 | 360.9 $\pm$ 4.5                   |
| Transformer (11L, matched)          | 45.2M  | 1.0e−4 | 352.5 $\pm$ 2.4                   |
| Per-block carrier (side-channel)    | 46.2M  | 1.0e−4 | 329.3 $\pm$ 3.6                   |
| <b>Spin-dominant (spin as core)</b> | 37.3M  | 6.7e−3 | <b>115.7 <math>\pm</math> 1.8</b> |

schedule-sensitive. The snapshot here (about  $3.1\times$ ) differs from the 21% above because budget, schedule, and learning rate differ across single-snapshot runs. To settle the worry that the transformer simply has not trained yet at the short budget, we ran both architectures to convergence on an identical schedule and batch stream (Fig. 4). The result goes the other way: the transformer descends and then plateaus (about 222) while the spin-dominant model descends to about 56, so the gap widens with training, not narrows. We also note the two architectures prefer very different learning rates (6.7e−3 vs 1.0e−4, a 67 $\times$  difference), consistent with state-space models tolerating higher rates.

#### D. Generalization: at the tested budget, placement, not phase

A natural objection is that “placement matters” might be specific to this diagonal-complex carrier rather than a property of recurrent mixers in general. We test it directly: we replace the spin carrier with a diagonal-real recurrence ( $h_t = \lambda \odot h_{t-1} + b_t$ , real  $\lambda \in (0, 1)$ , no phase), a different recurrence family, placed identically as the core, under the same multi-seed protocol (Table IV).

The real, no-phase recurrence as the core beats the attention baseline by 3.5 $\times$ , as much as the complex carrier (3.1 $\times$ ), with fewer parameters. At the tested budget, the placement effect generalizes to a different recurrence family; it is not an artifact of the complex/phase representation. Because the real recurrence here outperforms the complex one at this short budget, we claim only that placement (a recurrence as the core mixer) is what carries the computation at the tested scale and budget; whether the complex phase helps at larger scale or longer contexts is an open question this experiment does not settle.

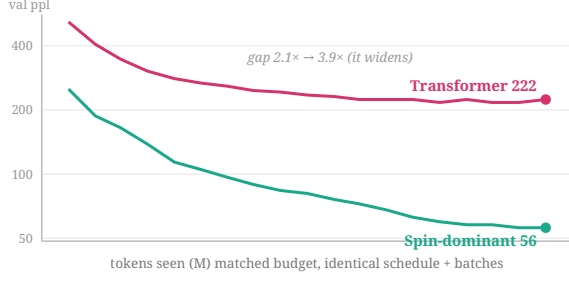


Fig. 4. Learning curves at matched 37M, identical schedule and batches, each at its range-test learning rate, to 117M tokens. Both converge, but the transformer plateaus at about 222 while the spin-dominant model descends to about 56, so the gap widens through training ( $2.1\times \rightarrow 3.9\times$ ). This answers the objection that the short-budget gap is just “the transformer hasn’t trained yet.” (Small-scale, single-seed.)

TABLE IV

GENERALIZATION CONTROL: A REAL, NO-PHASE RECURRENCE IN THE SAME CORE PLACEMENT (3 SEEDS, RANGE-TEST LR, MATCHED 30M-TOKEN BUDGET).

| Core token-mixer                | Params | Val ppl         | vs attn     |
|---------------------------------|--------|-----------------|-------------|
| Attention (transformer)         | 35.7M  | $360.9 \pm 4.5$ |             |
| Complex spin carrier            | 37.3M  | $115.7 \pm 1.8$ | $3.1\times$ |
| <b>Real diagonal (no phase)</b> | 34.1M  | $102.2 \pm 2.1$ | $3.5\times$ |

#### E. At scale: the spin carrier is the load-bearing core (284M)

We trained the intended design at real scale for the first time. A 284M model ( $d = 768$ , 20 layers, `mlp_mult=9`, carrier as mixer in 15 of 20 blocks, attention every 4th) trained to step 418,000 on a 176.6B-token chain-of-thought corpus, reaching best validation perplexity 24.54 (about 24 tokens/param; we report this as a description of the run, not a compute-optimality claim, since the corpus is deliberately reasoning-heavy). In the spin-dominant configuration the carrier is intra-block with no cross-segment state to transplant, so the equivalent causal measure is direct ablation.

**The carrier carries essentially the whole computation.** Gating the 15 SpinBlocks’ carrier to about 0 sends validation perplexity from 25 to 7716 ( $306\times$  worse); zeroing the 5 attention blocks’ output sends it from 25 to 85 ( $3.4\times$  worse). At 284M the spin carrier is the load-bearing core and attention a minor helper, the exact inverse of the 1.4B instance whose carrier was vestigial.

#### V. A SELF-CALIBRATING ADAPTIVE CONTROLLER

**Status.** This section describes an architecture and its mechanisms, not a validated set of results. Where a mechanism is measurable it works (self-growth fires only on probe-confirmed saturation; episodic writes and reads function); where the flagship behavior does not yet hold at this scale we say so plainly: the honesty gate does not separate knowledge from confabulation at 284M (Section VI), and the no-forgetting claim is asserted from the replay mechanism, not yet from a measured retention curve.

Around the language model runs a controller that decides, for any input, whether to answer, abstain, learn, seek, or think,

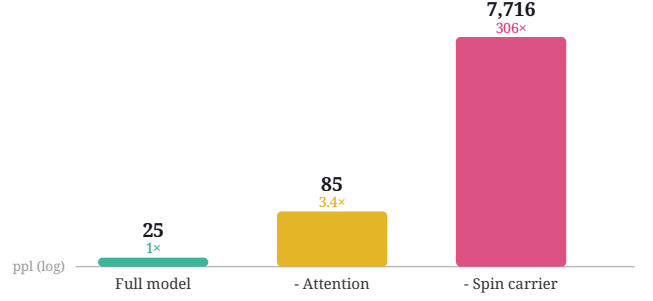


Fig. 5. Causal ablation on the 284M spin-dominant model (log scale). Zeroing the spin carrier sends validation perplexity from 25 to 7,716 ( $306\times$ ); zeroing attention only to 85 ( $3.4\times$ ). The carrier is the load-bearing core.

entirely from the model’s own signals, with no keyword rules. Four design commitments distinguish it.

#### A. No hand-set calibration

Every internal scale is derived from data and re-derived by a `recalibrate()` step as the model grows. Word importance is the self-information  $-\log p(\text{token})$  from the corpus. Honesty, novelty, and memory-match thresholds are set at the equal-error-rate crossover of two of the model’s own distributions, not hand-tuned. The training learning rate is found by a Smith range-test rather than seeded; the growth bar is the data budget (about 20 tokens/param [12]) and available memory.

#### B. Surprise-modulated continual learning

When taught a fact, the controller sets its own learning rate from its own surprise (prediction error relative to its familiarity boundary), learning hard on surprising input and gently on familiar input, and stopping once the fact ceases to surprise it. Teaching interleaves generative replay of the training distribution so a new fact does not erase old skills. A persistent in-weights episodic memory writes surprise-gated key $\rightarrow$ value traces of the model’s own activations, recalls them under an uncertainty gate, decays over time, and consolidates into the slow weights during a “sleep” step. It is a learned in-weights store, not retrieval-augmented generation.

##### Algorithm 1: Surprise-modulated continual update.

**inputs:** fact  $f$ , model  $M$ , replay corpus  $R$ , familiarity boundary  $\beta$ , base lr  
 $s \leftarrow \text{surprise}(M, f)$   
 $\eta \leftarrow \text{base\_lr} \cdot \text{clamp}(s/\beta, 0.15, 3)$   
 $A \leftarrow \{(x, \arg \max M(x)) : x \sim R\}$  (self-anchors)  
**repeat**  
 $L \leftarrow \text{CE}(M, f) + \lambda_r \text{CE}(M, x \sim R) + \lambda_a \text{CE}(M, A)$   
 $M \leftarrow \text{step}(M, \nabla L, \eta)$   
**until**  $\text{surprise}(M, f) < 0.4\beta$

#### C. Introspection and self-growth

An introspection layer adds a confidence self-estimate, step-by-step deliberation, and idle self-generation. Growth is function-preserving and fires only on probe-confirmed saturation: the model spawns a depth-grown and a width-grown copy, trains each briefly, and commits to growth only if

validation loss drops past the validation noise. This rule is confirmed in production: a spin-dominant run grew itself 24M (4 layers)  $\rightarrow$  27.3M (5 layers) on confirmed saturation, then stopped at the data budget, correcting the earlier over-growth.

## VI. EXPERIMENTS AND RESULTS

### A. Trained instances

The diagonal-complex carrier trains stably from 21M to 1.4B. Two instances anchor the analysis: the self-grown 1.4B single-carrier model (val ppl 17.35) is the attention-dominant “wrong build,” bigger but with a causally-negligible carrier; the 284M spin-dominant model (val ppl 24.54) is the corrected build,  $5\times$  smaller, with a provably load-bearing carrier. The training corpus is 176.6B tokens: 158.6B of base text plus about 18B of appended chain-of-thought reasoning. The base text is Cosmopedia, Wikipedia, and FineWeb-Edu, and the reasoning split is drawn from public, openly-licensed datasets, so the corpus is reproducible.

### B. Capability and honesty of the 284M model

A greedy capability battery on the 284M shows a clear jump from a 27M laptop model (27M in parentheses): arithmetic 3/3 (0/3), knowledge 2/3 (0/3), logic 2/2, theory-of-mind 2/2 (1/2), code 1/2, in-context binding 1/2; the still-weak axis is multi-step 0/2, counterfactual 0/2, causal 0/2. World knowledge, single-step arithmetic, and theory-of-mind emerged at this scale; multi-step composition is the next axis.

**An honest negative on honesty.** The controller answers only when its sampled answers agree on a content token. At 284M the signal does not cleanly separate knowledge from confident confabulation: a known fact and two unknowables score 0.53 / 0.53 / 0.67, overlapping. The model confabulates consistently. The gate errs toward “I don’t know” rather than guessing; reliable honesty needs scale or a redesigned signal.

### C. Zero-shot public benchmarks

For external comparability we evaluate the 284M model zero-shot on standard public sets. Because the model uses a custom 16K-token vocabulary, we report word-normalized perplexity and log-likelihood multiple-choice accuracy, both tokenizer-independent (Table V). The scores are honest and modest, above chance on most tasks and at chance on the hardest (ARC-Challenge), the expected profile of a small model undertrained for its size.

### D. Comparison with open models of similar size

We ran four open models through the identical harness (Table VI). On ARC-Easy our 284M (34.7) matches or beats GPT-2 (34.1) and Cerebras-256M (32.7), trailing only the larger GPT-2-medium; on HellaSwag it edges the similar-size models; ARC-Challenge is at chance for all four. On PIQA and especially LAMBADA our model is clearly behind, as LAMBADA rewards fluent general language modeling, which

TABLE V  
ZERO-SHOT PUBLIC BENCHMARKS (1,000-EXAMPLE SUBSAMPLES;  
ACC\_NORM = LENGTH-NORMALIZED LOG-LIKELIHOOD).

| Benchmark     | Metric                  | 284M         | Chance |
|---------------|-------------------------|--------------|--------|
| WikiText-2    | word ppl (lower better) | 157.9        |        |
| LAMBADA       | accuracy                | 19.0%        |        |
| HellaSwag     | acc_norm                | 27.9%        | 25%    |
| ARC-Easy      | acc_norm                | <b>34.7%</b> | 25%    |
| ARC-Challenge | acc_norm                | 24.0%        | 25%    |
| PIQA          | accuracy                | <b>59.5%</b> | 50%    |

a reasoning-heavy, undertrained model does least well. We include this not to claim a competitive small model, but to benchmark our numbers honestly. A 565M continuation confirms the trend: it improves on most tasks, and LAMBADA gains most (19.0  $\rightarrow$  21.7), as expected for an undertrained model.

### E. Progression across grown sizes

The grown models come from function-preserving growth followed by continued training. We tabulate every metric across the sizes (Table VII, Fig. 6). Two readings. (i) Growth and training compound cleanly: perplexity falls (24.5  $\rightarrow$  22.0  $\rightarrow$  21.97  $\rightarrow$  20.1), and, the strongest signal, the carrier becomes more causally dominant with scale, ablating it costs  $306\times$  at 284M,  $1081\times$  at 706M, and  $1886\times$  at 1B. The design is not a small-scale artifact; its core property strengthens. (ii) The continual long-context training pays off: at 565M the cross-window carry hurt (the model had never been trained with it); the 706M, trained with the curriculum mixed-window schedule, uses it, next-window perplexity now improving with the carry (21.6 vs 22.3).

### F. A non-controlled external reference (illustrative only)

A fair objection is that placing a recurrence at the core might cost multi-hop reasoning specifically. We cannot settle this without a same-corpus matched transformer, which is outside our compute budget. For rough context only, we ran one published open transformer of similar size and token budget (Cerebras-GPT-1.3B) through the identical harness (Table VIII). This is a non-controlled comparison: tokenizer, corpus, optimizer, and recipe all differ, so we draw no architectural conclusion from it. One observation is at least consistent with a scale (not architecture) reading of the multi-hop weakness: on ARC-Challenge both models sit at chance (24.3 and 25.7), the reference transformer reasoning no better at this scale either. The reference is ahead on general-English fluency, exactly where its larger size, general-web corpus, and standard tokenizer would help.

## VII. LIMITATIONS

Every claim above carries a number; the open frontier is equally explicit. (i) *The decisive at-scale experiment is missing, and the headline rests on one seed.* The quality win (spin-dominant beats a matched transformer) is shown at 37M,

TABLE VI

ZERO-SHOT ACCURACY (%) VS OPEN MODELS OF SIMILAR SIZE, ONE HARNESS, 1,000-EXAMPLE SUBSAMPLES. COMPARABLE TO GPT-2 (124M) AND CEREBRAS-256M, BEHIND THE LARGER GPT-2-MEDIUM; ARC-CHALLENGE SITS AT CHANCE FOR ALL FOUR MODELS.

| Model                   | Params | PIQA | ARC-Easy    | ARC-Challenge | HellaSwag | LAMBADA |
|-------------------------|--------|------|-------------|---------------|-----------|---------|
| <b>Pragnosia (ours)</b> | 284M   | 59.5 | <b>34.7</b> | 24.0          | 27.9      | 19.0    |
| <b>Pragnosia (ours)</b> | 565M   | 60.7 | 34.1        | 24.9          | 29.3      | 21.7    |
| GPT-2                   | 124M   | 63.3 | 34.1        | 23.1          | 27.2      | 32.2    |
| Cerebras-GPT-256M       | 256M   | 62.1 | 32.7        | 22.1          | 26.3      | 29.5    |
| GPT-2-medium            | 355M   | 66.5 | 37.5        | 23.3          | 36.8      | 42.8    |

TABLE VII

PROGRESSION OF THE SPIN-DOMINANT MODEL ACROSS GROWN SIZES (ALL THROUGH THE SAME HARNESSES). NEARLY EVERY METRIC IMPROVES OR HOLDS; LAMBADA GAINS MOST, AND THE CROSS-WINDOW CARRY FLIPS FROM A LIABILITY TO A BENEFIT ONCE THE MODEL IS TRAINED WITH IT.

| Metric                                  | 284M     | 565M      | 706M                 | 1B          |
|-----------------------------------------|----------|-----------|----------------------|-------------|
| Architecture ( $d$ / layers / mlp_mult) | 768/20/9 | 768/24/17 | 768/24/22            | 768/29/27   |
| Validation perplexity                   | 24.54    | 22.0      | <b>21.97</b>         | <b>20.1</b> |
| Carrier ablation ( $\times$ worse)      | 306      |           | <b>767</b>           | <b>1886</b> |
| Attention ablation ( $\times$ worse)    | 3.4      |           | 3.8                  | 4.6         |
| PIQA                                    | 59.5     | 60.7      | <b>61.8</b>          | 63.3        |
| ARC-Easy                                | 34.7     | 34.1      | <b>35.3</b>          | 37.2        |
| ARC-Challenge (chance)                  | 24.0     | 24.9      | 23.7                 | 24.3        |
| HellaSwag                               | 27.9     | 29.3      | <b>29.4</b>          | 29.6        |
| LAMBADA                                 | 19.0     | 21.7      | <b>24.4</b>          | 25.3        |
| Multi-hop battery (greedy)              | 2/10     | 3/10      | <b>4/11</b>          | 4/11        |
| Cross-window carry                      |          | hurts     | helps (21.6 vs 22.3) | helps       |

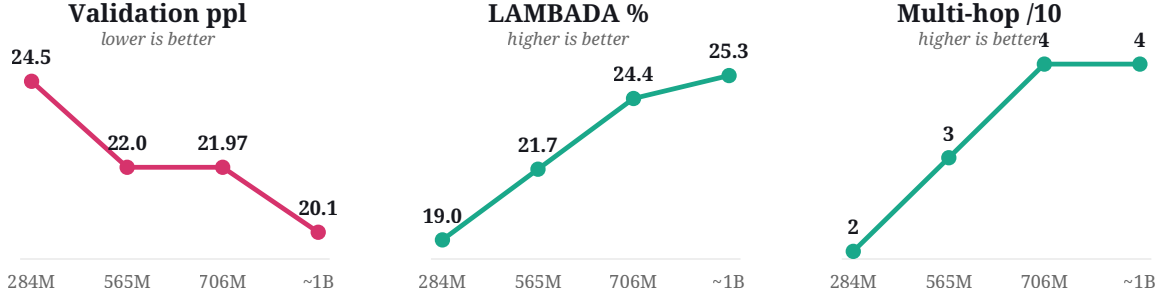


Fig. 6. Progression across the four grown sizes. Validation perplexity falls, and LAMBADA and the greedy multi-hop battery both rise monotonically, the design improving consistently as it grows and trains, with the largest gains on the tokens-limited axes.

three seeds; the causal-dominance property (the carrier is load-bearing) is shown at 284M, one seed. These are not the same claim, and the  $306\times$  number shows the carrier is used, not that the design wins at scale. The single highest-value experiment, an attention-only transformer at 284M with the same corpus and budget, is the crux, and it is still to run. The causal-dominance property itself does not need a baseline, and it persists and strengthens across scale: carrier ablation costs  $306\times$  at 284M,  $1081\times$  at 706M, and  $1886\times$  at 1B. A matched transformer at 1B is outside our compute budget and we do not claim it. (ii) *The models are undertrained for their grown size, and we treat this as a claim to be falsified, not a shield.* Single-step reasoning, arithmetic, and fact recall are real; multi-hop composition is weak. The falsifiable measurement is the full token-vs-capability curve on the multi-hop axis;

if more tokens do not move it, the undertraining account is wrong. (iii) *Honesty is not solved.* The consistency-based gate abstains on the clearly-unknowable but cannot separate knowledge from confident confabulation at 284M; it needs scale or a redesigned signal. (iv) Like all small models it can be confidently wrong on familiar-but-incorrect facts. (v) *Long context: aggregate use is realized; sharp retrieval is not.* A model trained at a fixed 256-token window never learns to use a cross-window carried state; at 565M, feeding one hurt. After implementing cross-window state-threading and training with it, the 706M carry helps (next-window ppl 21.6 vs 22.3), and longer effective context lowers perplexity. We verify the inference claim directly on the 1B: greedy and sampled generation both run at a constant 27 tokens/s from 256 to 8000 tokens. Coherence over that span is decoding-

TABLE VIII

OUR 1B (SPIN-DOMINANT) VS A ROUGHLY TOKEN-MATCHED TRANSFORMER (ILLUSTRATIVE, NON-CONTROLLED). ON THE MULTI-HOP TASK (ARC-CHALLENGE) BOTH SIT AT CHANCE; THE TRANSFORMER’S LEAD IS CONCENTRATED ON GENERAL-ENGLISH FLUENCY, WHERE OUR CUSTOM CORPUS AND TOKENIZER ARE CONFOUNDS.

| Model                           | Params | PIQA | ARC-Easy | ARC-Challenge | HellaSwag | LAMBADA |
|---------------------------------|--------|------|----------|---------------|-----------|---------|
| <b>Pragnosia (ours, spin)</b>   | 1022M  | 63.3 | 37.2     | 24.3          | 29.6      | 25.3    |
| Cerebras-GPT-1.3B (transformer) | 1316M  | 67.1 | 38.9     | 25.7          | 35.9      | 47.0    |

dependent, not architectural: greedy repetition-loops by about 1k tokens, whereas temperature sampling stays on-theme to 8k. What is not yet there is sharp long-range retrieval: the damped carry provides gist, not verbatim memory.

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### VIII. CONCLUSION

Pragnosia is a transformer/state-space hybrid in which a diagonal-complex spin recurrence is the core token-mixer, wrapped in a self-calibrating controller for continual learning, a persistent in-weights episodic memory, an introspection layer, and self-governed growth, with no hand-set constants. The central, honest finding is about placement: a single spin carrier drifts to a causally-negligible side-channel as the model scales (0.028% of the loss at 1.4B), and a parameter-matched ablation shows the intended spin-dominant design is the most parameter-efficient (219 vs 279 ppl). At 284M, the first at-scale training of the intended build, ablating the carrier costs  $306\times$  in perplexity versus  $3.4\times$  for attention: the spin carrier is the load-bearing core, the exact inverse of the 1.4B drift. The remaining work is a matched transformer baseline and multi-seed replication at 1B, and training the models to their compute-optimal token budget. We release the full system, every measured number, and every limitation.

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